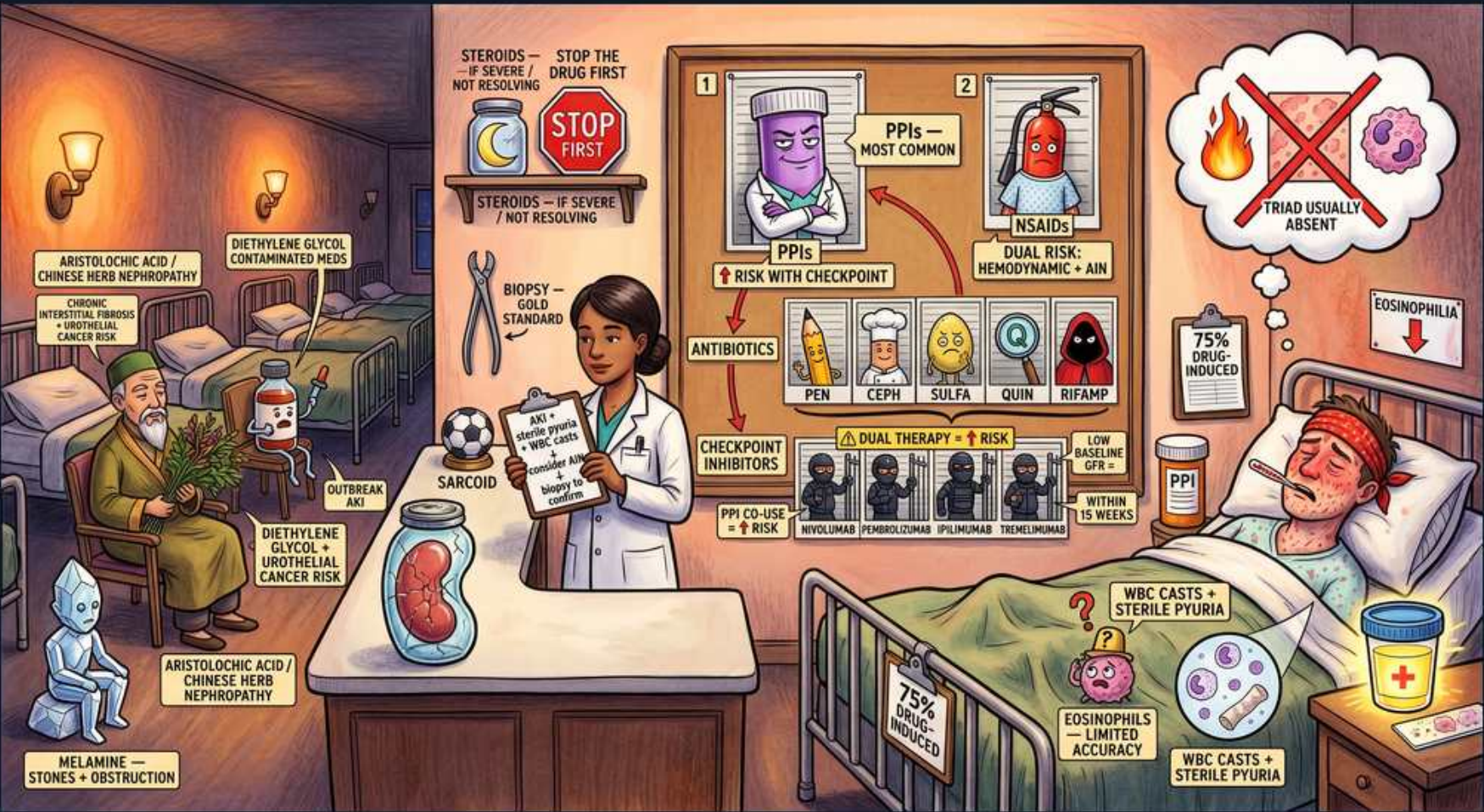


AKI — SCENE AKI-03: The AIN Sick Bay



1. AIN sick bay

WHAT YOU SEE

A patient lying ill in a hospital sick-bay bed at the corner of the ward — the bedridden allergic-reaction patient IS acute interstitial nephritis, the drug-triggered inflammation that lands the kidney in sick bay.

WHAT IT ENCODES

Acute interstitial nephritis (AIN) is a drug-induced, T-cell-mediated (type IV hypersensitivity) allergic inflammation of the renal interstitium and tubules, sparing glomeruli and vessels. It is a common cause of intrinsic AKI presenting as a subacute rise in creatinine days to weeks after starting a culprit drug. Pathology shows interstitial edema with a lymphocyte/eosinophil infiltrate.

BOARD TRAP

AIN is often mistaken for ATN because both are intrinsic AKI — the discriminator is the clinical context (new drug, allergic features, WBC casts/eosinophiluria) versus ATN's ischemia/toxin history and muddy-brown granular casts.

2. PPIs most common

WHAT YOU SEE

Purple PPI chef '1 - PPIs MOST COMMON', risk with checkpoint

WHAT IT ENCODES

Proton-pump inhibitors (omeprazole, pantoprazole) are now the most common drug cause of AIN, overtaking antibiotics, partly because of how widely they are used. Onset is typically insidious (weeks to months) and allergic features are usually absent. PPI co-use also amplifies the AIN risk of checkpoint inhibitors. Mnemonic for AIN drugs: '5 P's' — Pee (diuretics), Pain-free (NSAIDs), Penicillins/cephalosporins, Proton-pump inhibitors, rifamPin.

BOARD TRAP

Because PPIs are so common and benign-seeming, students overlook them as a cause — WRONG. In a patient on a chronic PPI with unexplained subacute AKI and bland-to-pyuric urine, stop the PPI before pursuing rarer causes.

3. NSAIDs dual risk

WHAT YOU SEE

NSAIDs fire extinguisher '2', 'DUAL RISK hemodynamic + AIN'

WHAT IT ENCODES

NSAIDs cause renal injury by TWO mechanisms: (1) hemodynamic prerenal AKI from prostaglandin inhibition causing afferent arteriolar vasoconstriction, and (2) AIN, often combined with minimal-change disease producing nephrotic-range proteinuria. NSAID-induced AIN characteristically has a LONGER latency (months) and frequently lacks the classic allergic triad.

BOARD TRAP

Seeing nephrotic-range proteinuria, a student excludes AIN and calls it primary glomerular disease — WRONG. NSAID-AIN uniquely pairs interstitial nephritis WITH heavy proteinuria (minimal change), a combination tested specifically. Also don't reflexively blame only hemodynamics; biopsy may show AIN.

4. Antibiotics

WHAT YOU SEE

ANTIBIOTICS row: PEN, CEPH, SULFA, QUIN, RIFAMP

WHAT IT ENCODES

Antibiotic AIN culprits: penicillins (classically methicillin), cephalosporins, sulfonamides, fluoroquinolones, and rifampin. Beta-lactam AIN more often shows the allergic triad and rapid onset. Rifampin can cause an acute, sometimes rapidly progressive AIN, especially with intermittent dosing.

BOARD TRAP

Sulfa drugs cause AIN AND can crystallize in tubules — don't confuse antibiotic AIN with crystal-induced AKI (acyclovir, methotrexate, sulfonamide crystalluria); urine microscopy distinguishes WBC casts/eosinophils from crystals.

5. Checkpoint inhibitors

WHAT YOU SEE

CHECKPOINT INHIBITORS, nivolumab/pembrolizumab/ipilimumab figures, 'PPI co-use up risk'

WHAT IT ENCODES

Immune checkpoint inhibitors (PD-1: nivolumab, pembrolizumab; CTLA-4: ipilimumab; tremelimumab) cause immune-related AIN, typically with delayed onset (median ~14–15 weeks, later than classic drug AIN) and a low GFR at diagnosis. Risk is increased by concurrent PPI use. Management is drug-hold plus high-dose corticosteroids, and it is one AIN scenario where steroids are central to recovery.

BOARD TRAP

Attributing a checkpoint-inhibitor patient's AKI to ATN/prerenal causes delays steroids — WRONG. The discriminator is the immunotherapy exposure with subacute AKI; biopsy confirms AIN and steroids are indicated, unlike most other AKI where steroids have no role.

6. Triad usually absent

WHAT YOU SEE

Thought bubble: red X over fever/rash/eosinophil

WHAT IT ENCODES

The classic AIN triad of fever, maculopapular rash, and peripheral eosinophilia is present in only a MINORITY of patients (<10–15% have all three). Its absence does NOT rule out AIN. Modern PPI- and NSAID-induced AIN are especially likely to lack the triad.

BOARD TRAP

A student excludes AIN because the patient has no fever/rash/eosinophilia — WRONG, that is the single most-tested AIN pitfall. The triad is specific but very insensitive; biopsy, not the triad, confirms or excludes AIN.

7. Eosinophilia

WHAT YOU SEE

EOSINOPHILIA down arrow card near patient

WHAT IT ENCODES

Peripheral eosinophilia and urine eosinophils (Hansel/Wright stain) support AIN but are neither sensitive nor specific. Urine eosinophils also occur in atheroembolic disease, pyelonephritis, and others, so they cannot confirm or exclude AIN.

BOARD TRAP

Treating absence of eosinophilia/eosinophiluria as ruling out AIN, or presence as proving it, are both wrong — these tests have been largely abandoned for diagnosis. Biopsy remains the gold standard.

8. WBC casts / sterile pyuria

WHAT YOU SEE

WBC CASTS + STERILE PYURIA cards, eosinophils

WHAT IT ENCODES

AIN urinalysis classically shows white-cell casts, sterile pyuria (pyuria with negative culture), microscopic hematuria, and sometimes eosinophiluria. Proteinuria is usually mild EXCEPT in NSAID-AIN where it can be nephrotic-range.

BOARD TRAP

WBC casts are NOT specific to AIN — they also appear in pyelonephritis. The discriminator from infection is the STERILE culture; the discriminator from ATN is that ATN shows muddy-brown GRANULAR casts and renal tubular epithelial cells, not WBC casts.

9. 75% drug-induced

WHAT YOU SEE

'75% DRUG-INDUCED' card

WHAT IT ENCODES

Roughly 70–75% of AIN cases are drug-induced. The remainder stem from infections (Legionella, leptospirosis, CMV, etc.), autoimmune/systemic disease (SLE, Sjögren, IgG4-related disease, TINU syndrome), and sarcoidosis (granulomatous interstitial nephritis).

BOARD TRAP

Assuming AIN is always drug-related delays the workup in the ~25% with infectious or systemic causes — in a young patient with AIN plus uveitis, think TINU (tubulointerstitial nephritis and uveitis) syndrome rather than a drug.

10. Stop the drug first

WHAT YOU SEE

STOP FIRST sign, 'steroids if severe / not resolving'

WHAT IT ENCODES

First and most important step in drug-induced AIN is to STOP the offending drug; many patients recover with this alone. Corticosteroids (e.g., prednisone ~1 mg/kg/day) are added if renal function is severe, fails to improve after drug withdrawal, or biopsy shows extensive inflammation — and earlier treatment correlates with better recovery.

BOARD TRAP

Starting steroids before (or instead of) withdrawing the culprit drug is the wrong sequence — removing the trigger comes first. Continuing the drug while giving steroids undermines recovery.

11. Biopsy gold standard

WHAT YOU SEE

BIOPSY GOLD STANDARD, clinician with card, sarcoid kidney

WHAT IT ENCODES

Kidney biopsy is the diagnostic gold standard for AIN, showing interstitial edema and a lymphocyte/eosinophil infiltrate; non-caseating granulomas suggest sarcoid or drug-induced granulomatous interstitial nephritis. Biopsy is pursued when the diagnosis is uncertain, AKI is severe, or there is no recovery after stopping the drug.

BOARD TRAP

Relying on urine eosinophils or the clinical triad to confirm AIN instead of biopsy is wrong when the diagnosis or treatment decision is unclear — those tests are too inaccurate; histology is definitive.

12. Aristolochic acid

WHAT YOU SEE

ARISTOLOCHIC ACID / Chinese herb nephropathy, herbalist

WHAT IT ENCODES

Aristolochic acid, found in certain Chinese/herbal slimming remedies and as a contaminant of wheat (Balkan endemic nephropathy), causes a CHRONIC tubulointerstitial fibrosis with progressive CKD and a markedly elevated risk of upper-tract urothelial carcinoma. This is a chronic interstitial nephritis, distinct from acute drug-AIN.

BOARD TRAP

A student lumps aristolochic acid with acute AIN — WRONG; it causes chronic interstitial FIBROSIS plus urothelial cancer, so the board point is to screen for malignancy, not just stop a drug.

13. Diethylene/melamine

WHAT YOU SEE

DIETHYLENE GLYCOL contaminated meds, MELAMINE stones figure

WHAT IT ENCODES

Toxic contaminants causing outbreak AKI: diethylene glycol (a deadly adulterant in counterfeit cough syrups/medications, causing tubular necrosis and neuropathy) and melamine (added to fake-up protein content in food/formula, causing urinary stones and obstructive uropathy, classically in infants).

BOARD TRAP

These are easily confused with each other — diethylene glycol = contaminated MEDICATIONS causing direct toxic AKI, whereas melamine = contaminated FOOD/formula causing STONES and obstruction. Don't confuse diethylene glycol (medication contaminant) with ETHYLENE glycol (antifreeze, oxalate crystals).